

# Unsupervised learning

## Supervised Learning

- Data: **labeled** data, feature-response pairs  $(x, y)$
- Goal: to **learn a mapping** from  $x$  to  $y$
- Problem type:
  - Classification
  - Regression

A supervised learning model can still be used for a confirmatory or inferential goal, such as testing a coefficient or estimating a causal effect; that is a different question about the purpose of the analysis.

## Unsupervised Learning

- Data: **unlabeled** data, features  $x$
- Goal: to uncover **hidden structure**, patterns, clusters, ...
- Typical tasks
  - Clustering analysis: cluster features or observations into groups with common traits
  - Dimension reduction: transform data from high-dimensional space to a low-dimensional space, while preserving as much information as possible
  - .....

## Reinforcement Learning

- Data: a sequence of **state-action pairs** and a **cumulative reward**
- Goal: to learn a **policy (a mapping from state to action)** that maximizes the **cumulative reward**.
- Typical applications:
  - Game-playing (e.g., AlphaGo, Dota2),
  - Robotics control
  - Autopilot, FSD
  - .....

## Quick Summary

| Aspect         | Supervised Learning                    | Unsupervised Learning                   | Reinforcement Learning                         |
|----------------|----------------------------------------|-----------------------------------------|------------------------------------------------|
| Goal           | Learn a <b>mapping</b> from $x$ to $y$ | Discover <b>hidden structure</b> in $x$ | Learn a <b>policy</b> that maximizes reward    |
| Data           | <b>Labeled data</b>                    | <b>Unlabeled data</b>                   | State-action pairs, and <b>delayed rewards</b> |
| Typical output | Classification or regression model     | Clusters, low-dimensional embeddings    | Policy                                         |

## Unsupervised Learning: Clustering Analysis

- Popular techniques in analysis of complex data (e.g., genomics data)
- Aims to detecting **homogeneous subgroups** among the observations or the features
- Identities of the learned subgroups are unknown, but can often lead to meaningful interpretation after linked with additional information.
- We can cluster observations or features
  - observations: clustering cancer samples to find similar cancer sub-types
  - features: clustering genes based on similar functions

## Meaningful clusters

Without the groundtruth labels, how do we claim that our clusters results are meaningful?

[Link to a clustering playground](#)

## Similarity and dissimilarity measures

Often we have to come up with numeric metrics to measure the similarity or dissimilarity between observations.

### Similarity measures

- Pearson correlation
- Spearman correlation
- Kendall's  $\tau$
- Cosine similarity ([scikit-learn pairwise metrics](#))
- Mutual information for comparing labelings ([scikit-learn](#))

### • Dissimilarity measures

- Euclidean distance
- Manhattan distance
- Hamming / Jaccard distances for binary or set-like data ([SciPy distances](#))
- Wasserstein distance for distributions ([SciPy](#))
- Bottleneck or Wasserstein distances between persistence diagrams in topological data analysis ([GUDHI bottleneck](#), [GUDHI Wasserstein](#))

The key question is not "which measure is best?" but "what does it mean for two observations to be close in this problem?"

## Similarity measures

When the features are numeric:

- Pearson correlation: the usual correlation coefficient, measures **linear** association between  $X_i$  and  $X_j$ .
- Spearman correlation: Pearson correlation applied to **ranked** observations.
- Kendall's *au*: compares the ordering among pairs of observations.
- Cosine similarity: compares the **angle** between two feature vectors; useful when magnitude is less important than direction.

When the features are not numeric:

- Jaccard similarity: overlap between two sets, such as shared words, tags, or clicked items.
- Hamming similarity/distance: how many categorical or binary entries match.
- Edit distance: how many insertions, deletions, or substitutions separate two strings.
- Graph similarity: compare nodes by shared neighbors, paths, or graph embeddings.
- Mixed-type similarity: combine numeric, categorical, and binary features after deciding how each type should count.

## Dissimilarity measure

- **Euclidean distance**: sum of squared difference between feature values ( $\ell_2$ -norm)

$$d_E(i, j) = \left[ \sum_{l=1}^p (X_{i,l} - X_{j,l})^2 \right]^{1/2}$$

- **Manhattan distance**: sum of absolute difference between feature values ( $\ell_1$ -norm)

$$d_M(i, j) = \sum_{l=1}^p |X_{i,l} - X_{j,l}|$$

- .....

Checkout [this example](#).

## Common Clustering Methods

- **Hierarchical clustering**
  - Sequence of solutions organized in a hierarchical tree structure, called the dendrogram
  - Maintain a hierarchical structure  
When the number of clusters decrease, observations in the same cluster stays in the same cluster
- **k-Means clustering**
  - One of the most popular clustering method
  - Computationally efficient
  - No hierarchy among clusters  
When  $k$  decreases, observations in the same clusters might be reassigned to different clusters
- And [many other methods](#)

## Dimension Reduction

- Goal: to transform data from a high-dimensional space to a low-dimensional space, without losing too much information
- Versatile tools for
  - Exploratory data analysis (e.g., tSNE)
  - Data processing for downstream analysis (e.g., principal component regression)
  - Uncover true latent pattern (e.g., intrinsic dimension of neural activities)
- Typical methods
  - **Principal Component Analysis** (PCA)
  - **t-distributed Stochastic Neighbor Embedding** (tSNE)